# Test 8 – Grade 9

**Name: _____**
**Grade / Class: 9**

---

## Section A: English Language & Literature (Questions 1–15)

1. **Read the passage: *"Maya explored the city library, cataloged books on ancient civilizations, and interviewed the librarian about rare manuscripts."***
   **Question: What activity was Maya mainly involved in at the library?**
2. **Identify two sources of information Maya used during her research.**
3. **Explain the importance of cataloging books.**
4. **Use the word *interview* in a complete sentence.**
5. **Change into passive voice: *"Maya interviewed the librarian."***
6. **Correct the sentence: *"Each of the students have submitted their essays."***
7. **Identify the underlined word in: *"The librarian explained patiently."* (part of speech)**
8. **Write a formal email to a publisher requesting permission to cite a book in your project (80–100 words).**
9. **Compose a short poem on *"A Rainy Afternoon"* (50–60 words).**
10. **Define the literary device *metaphor* and give an original example.**
11. **Differentiate between subjective and objective writing with examples.**
12. **Summarize the passage in 50 words.**
13. **Turn this sentence into a question: *"The team completed the survey efficiently."***
14. **Write a paragraph describing a cultural festival you recently attended (60–80 words).**
15. **Explain the theme of this quote: *"Knowledge is the key to freedom."***

---

## Section B: Mathematics (Questions 16–35)

16. **Solve: $3x - 7 = 14$**
17. **Factorize: $x^2 + 11x + 24$**
18. **Simplify: $(4x^2y)(3xy^3)$**
19. **Solve the system:**
    **$x + 2y = 7$**
    **$3x - y = 8$**
20. **Calculate the area of a circle with radius 7 cm.**
21. **Solve: $5(2x - 3) = 3x + 9$**
22. **Expand: $(x + 5)(x - 2)$**
23. **Find the LCM of 12 and 18.**

**24. A train travels 480 km in 6 hours. Find its average speed.**
**25. Solve: $7y/3 + 4 = 9$**
**26. Find the roots of $x^2 - 7x + 12 = 0$.**
**27. Find the perimeter of a square with side length 9 cm.**
**28. Simplify: $(3a^2b^2)(2ab^3)$**
**29. A bag contains 6 red, 5 blue, and 9 yellow balls. Probability of picking a blue ball?**
**30. Solve: $4(x - 5) + 6 = 2x + 14$**
**31. Convert 0.625 into a fraction.**
**32. A polygon has a sum of interior angles 1260°. Find the number of sides.**
**33. Solve: $8(x - 3) = 4x + 12$**
**34. The ratio of cats to dogs in a shelter is 5:6. If there are 25 cats, how many dogs?**
**35. Simplify: $(6x^3y^2) \div (2xy)$**

---

## Section C: Physics (Questions 36–50)

**36. State Newton's First Law of Motion.**
**37. A 10 N force acts on a 2 kg object. Find the acceleration.**
**38. Differentiate between instantaneous speed and average speed.**
**39. A car decelerates from 100 m/s to 40 m/s in 12 seconds. Find the deceleration.**
**40. Define kinetic energy and give an example.**
**41. A 3 kg object is raised to 8 m. Calculate its potential energy ($g = 10\ m/s^2$).**
**42. Explain conservation of energy with a real-life example.**
**43. Define work and write its SI unit.**
**44. A force of 25 N moves a box 4 m. Calculate the work done.**
**45. Explain scalar and vector quantities with examples.**
**46. State Snell's Law.**
**47. Differentiate conduction, convection, and radiation.**
**48. A 10 Ω resistor has 5 A current. Find the voltage.**
**49. Define electrical resistance.**
**50. Calculate the energy consumed by a 150 W lamp used for 5 hours.**

---

## Section D: Chemistry (Questions 51–65)

**51. Define a salt and give an example.**
**52. Write the chemical formula for sodium bicarbonate.**
**53. Balance: $N_2 + H_2 \rightarrow NH_3$**
**54. Define an alkali and give an example.**
**55. Differentiate between homogeneous and heterogeneous mixtures.**
**56. Write electron configuration of carbon.**

57. **Define reduction and oxidation.**
58. **Products of acid + base reaction?**
59. **Classify freezing water as physical or chemical change.**
60. **Describe the significance of Mendeleev's periodic table.**
61. **Give formula for nitric acid.**
62. **Explain isotope with an example.**
63. **Difference between metals and metalloids.**
64. **Chemical reaction for combustion of methane.**
65. **Describe the pH scale and mention acids and bases.**

---

## Section E: Biology (Questions 66–80)

66. **Define respiration and give the equation.**
67. **Write the chemical formula for aerobic respiration.**
68. **Function of chloroplasts.**
69. **Compare plant and animal cells.**
70. **Describe human lungs structure.**
71. **What is diffusion? Provide an example.**
72. **Role of pancreas in digestion.**
73. **Two diseases caused by viruses.**
74. **Define ecosystem.**
75. **Difference between amphibians and reptiles.**
76. **Stages of human menstrual cycle.**
77. **Types of skeletal muscles.**
78. **Define innate and adaptive immunity.**
79. **Explain carbon cycle.**
80. **Role of stomata in plants.**

---

## Section F: Social Studies / History / Geography (Questions 81–90)

81. **Define federal government.**
82. **Three largest deserts in Africa.**
83. **Causes of the French Revolution.**
84. **Define global warming.**
85. **Capital of Canada.**
86. **Explain freedom of speech.**
87. **Major mountain ranges in Europe.**
88. **Effects of deforestation.**

**89. Name three ancient Mesopotamian civilizations.**
**90. Difference between equator and prime meridian.**

---

## Section G: Computer / ICT (Questions 91–95)

**91. Define algorithm.**
**92. Name two types of storage devices.**
**93. Functions of spreadsheet software.**
**94. Difference between analog and digital signals.**
**95. Define cloud computing.**

---

## Section H: Islamiyat / Religious Studies (Questions 96–100)

**96. List the six articles of faith in Islam.**
**97. Explain the importance of Fasting in Ramadan.**
**98. Define Hajj and its significance.**
**99. Who was Prophet Muhammad (PBUH) and why is he important?**
**100. Story of Prophet Musa (Moses) and the main lesson learned.**